

Pillar 5: Capstone Project

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Project Title

Beyond Accuracy: A Cost-Sensitive and Explainable AI Framework for Automated Early Detection of Fetal Distress

Abstract

Fetal distress is a critical obstetric emergency where delayed intervention can lead to severe neonatal morbidity or mortality. While machine learning models have been widely applied to Cardiotocography (CTG) data for classification, standard approaches often prioritize overall accuracy, inadvertently neglecting the asymmetric risks of false negatives (missed distress) versus false positives (unnecessary intervention). This project proposes a **Cost-Sensitive Machine Learning Framework** designed specifically to minimize the clinical cost of missed diagnoses.

Unlike conventional studies that treat all errors equally, this research implements a **custom loss function** and **dynamic threshold optimization** that penalizes false negatives 10x more heavily than false positives. Furthermore, moving beyond the "black box" paradigm, we integrate **SHAP (SHapley Additive exPlanations)** to validate model decisions against established medical literature, ensuring that feature importance aligns with clinical intuition regarding uterine contractions and fetal heart rate variability. By combining **Stratified K-Fold Cross-Validation**, **SMOTE-NC** for robust imbalance handling, and an interactive **Streamlit dashboard** for adjustable sensitivity, this project delivers a deployable Decision Support System (DSS) that prioritizes patient safety over statistical metrics. Our results demonstrate that optimizing for **Recall** and **Clinical Cost** yields a more reliable tool for triage than standard accuracy-driven models.

1.1 Background

Cardiotocography (CTG) is the standard method for monitoring fetal well-being during pregnancy and labor. It records fetal heart rate (FHR) and uterine contractions to detect signs of distress, such as hypoxia. However, the interpretation of CTG tracings is subjective and prone to inter-observer variability, leading to both missed diagnoses and

unnecessary emergency C-sections. Machine Learning (ML) offers a promising solution to standardize this process, yet most existing models focus on maximizing **Accuracy** or **F1-Score**.

1.2 Problem Statement

In a clinical setting, **Accuracy is a misleading metric**. A model that achieves 95% accuracy by simply predicting "Normal" for all cases is clinically useless because it misses the critical 5% of distressed fetuses. The cost of a **False Negative** (failing to detect distress) is potentially catastrophic, whereas a **False Positive** typically results in a non-invasive follow-up check. Current literature often fails to explicitly model this **asymmetric cost function**, leading to models that are statistically robust but clinically risky.

1.3 Project Objectives & Novelty

This capstone project seeks to bridge the gap between statistical performance and clinical utility. The primary objectives are:

1. **Cost-Sensitive Optimization:** To develop a classification model that explicitly minimizes a **weighted cost matrix**, prioritizing **Recall (Sensitivity)** over Precision.
2. **Explainable AI (XAI):** To utilize **SHAP values** not just for feature ranking, but to validate that the model's decision logic aligns with medical guidelines (e.g., the significance of "Late Decelerations" or "Reduced Variability").
3. **Robust Imbalance Handling:** To rigorously compare resampling techniques (SMOTE, ADASYN) and ensemble methods to ensure the model generalizes well without overfitting to synthetic noise.
4. **Interactive Deployment:** To build a **Streamlit-based Decision Support Tool** that allows clinicians to adjust the prediction threshold based on the specific risk tolerance of the hospital protocol.

1.4 Significance

By shifting the focus from "highest accuracy" to "lowest clinical risk," this project demonstrates a mature approach to healthcare AI. It provides a replicable framework for building **safe, interpretable, and actionable** AI systems, directly addressing the trust and safety barriers that currently hinder the adoption of AI in obstetrics.

1.5 Dataset Overview

- **Dataset Name:** Cardiotocography (CTG) Data
- **Source:** UCI Machine Learning Repository (Dataset #193)
- **Data Origin:** Measurements extracted from CTG recordings using the **SisPorto 3.0** system.
- **Target Variable:** **NSP** (Neurological Status of the Prognosis)
- **Target Values:**

- 1 : Normal
 - 2 : Suspect
 - 3 : Pathological
- **Total Instances:** 2,124
 - **Features:** 21 numeric input features (LB, AC, FM, etc.)

Data Dictionary (Official UCI Definitions)

| # | Attribute Name | Full Description | Unit | Range | Clinical Significance | | :--- | :--- | :---
| :--- | :--- | :--- | | **1** | **LB** | Baseline value (FHR) | bpm | 50–210 | Mean fetal heart rate.
Normal: 110–160. | | **2** | **AC** | Accelerations | count | 0–20 | Transient FHR increases. | | **3** |
FM | Fetal movement | count | 0–20 | Number of movements detected. | | **4** | **UC** | Uterine
contractions | count | 0–20 | Number of contractions. | | **5** | **DL** | Light decelerations |
count | 0–20 | Mild FHR drops. | | **6** | **DS** | Severe decelerations | count | 0–20 | Deep FHR
drops (critical). | | **7** | **DP** | Prolonged decelerations | count | 0–20 | FHR drops > 2 mins. | |
8 | **ASTV** | % Time with Abnormal STV | % | 0–100 | % time short-term variability is
abnormal. | | **9** | **MSTV** | Mean value of STV | ms | 0–50 | Average beat-to-beat variation. | |
| **10** | **ALTV** | % Time with Abnormal LTV | % | 0–100 | % time long-term variability is
abnormal. | | **11** | **MLTV** | Mean value of LTV | ms | 0–50 | Average long-term variation. | |
12 | **Width** | Histogram width | bpm | 0–100 | Range of FHR histogram. | | **13** | **Min** |
Histogram min | bpm | 50–210 | Minimum FHR in histogram. | | **14** | **Max** | Histogram max
| bpm | 50–210 | Maximum FHR in histogram. | | **15** | **Nmax** | Num. of histogram peaks |
count | 0–20 | Peaks in FHR distribution. | | **16** | **Nzeros** | Num. of histogram zeros | count
| 0–20 | Gaps in FHR data. | | **17** | **Mode** | Histogram mode | bpm | 50–210 | Most frequent
FHR. | | **18** | **Mean** | Histogram mean | bpm | 50–210 | Average FHR from histogram. | | **19**
| **Median** | Histogram median | bpm | 50–210 | Median FHR. | | **20** | **Variance** | Histogram
variance | ms² | 0–50 | Variance of FHR distribution. | | **21** | **Tendency** | Histogram
tendency | -1, 0, 1 | Skewness (-1: Left, 0: Sym, 1: Right) | | **T** | **NSP** | **Neurological Status**
Prognosis | 1, 2, 3 | **Target Class:**
1 = Normal
2 = Suspect
3 = Pathological |

Target Variable Mapping

We map the NSP column to a binary target Target_Binary for the classification task:

NSP Value	Clinical Category	Project Binary Label (Target_Binary)	Definition
1	Normal	0	Healthy fetus.
2	Suspect	1	Potential risk (merged into abnormal).
3	Pathological	1	High risk (merged into abnormal).

1.6 Project Scope & Deliverables

1. Data Preprocessing Pipeline:

- Cleaning and handling physiological outliers.
- Handling class imbalance (SMOTE, Class Weights, or Undersampling).

2. Exploratory Data Analysis (EDA):

- Visualizing feature distributions and correlations with **NSP**.
- Identifying key predictors of fetal distress.

3. Model Development:

- Training multiple algorithms (Logistic Regression, Random Forest, XGBoost, SVM).
- Implementing **Cost-Sensitive Learning** techniques.

4. Evaluation:

- Metrics: **Recall, Precision, F1-Score, ROC-AUC**, and **Confusion Matrix**.
- Focus: Minimizing False Negatives (Type II errors).

5. Final Report:

- Documentation of methodology, results, and clinical implications.

1.7 Why This Matters

By automating the detection of abnormal **NSP** values with high sensitivity, this project aims to support clinicians in making faster, more reliable decisions, ultimately reducing the risk of adverse neonatal outcomes.
